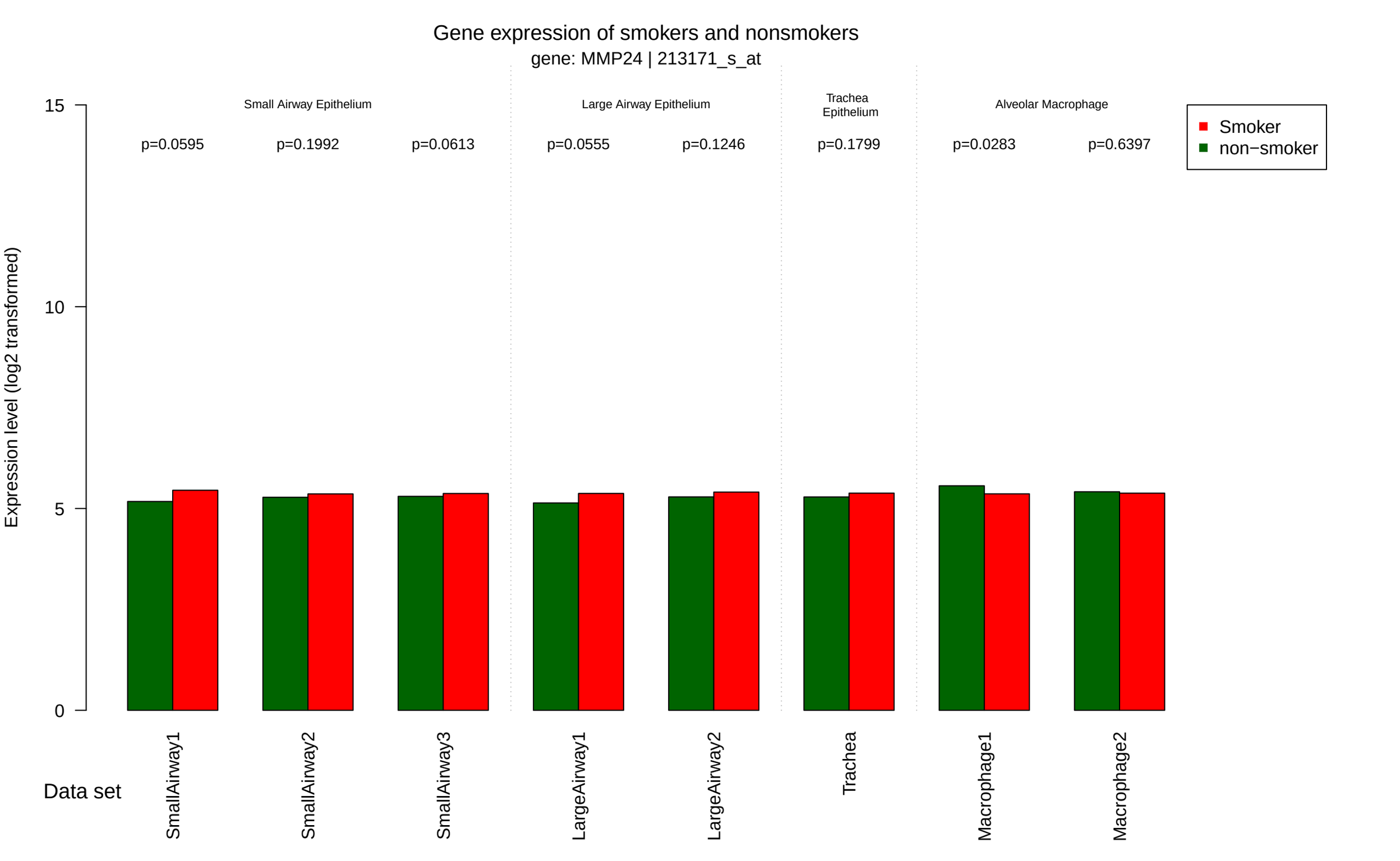
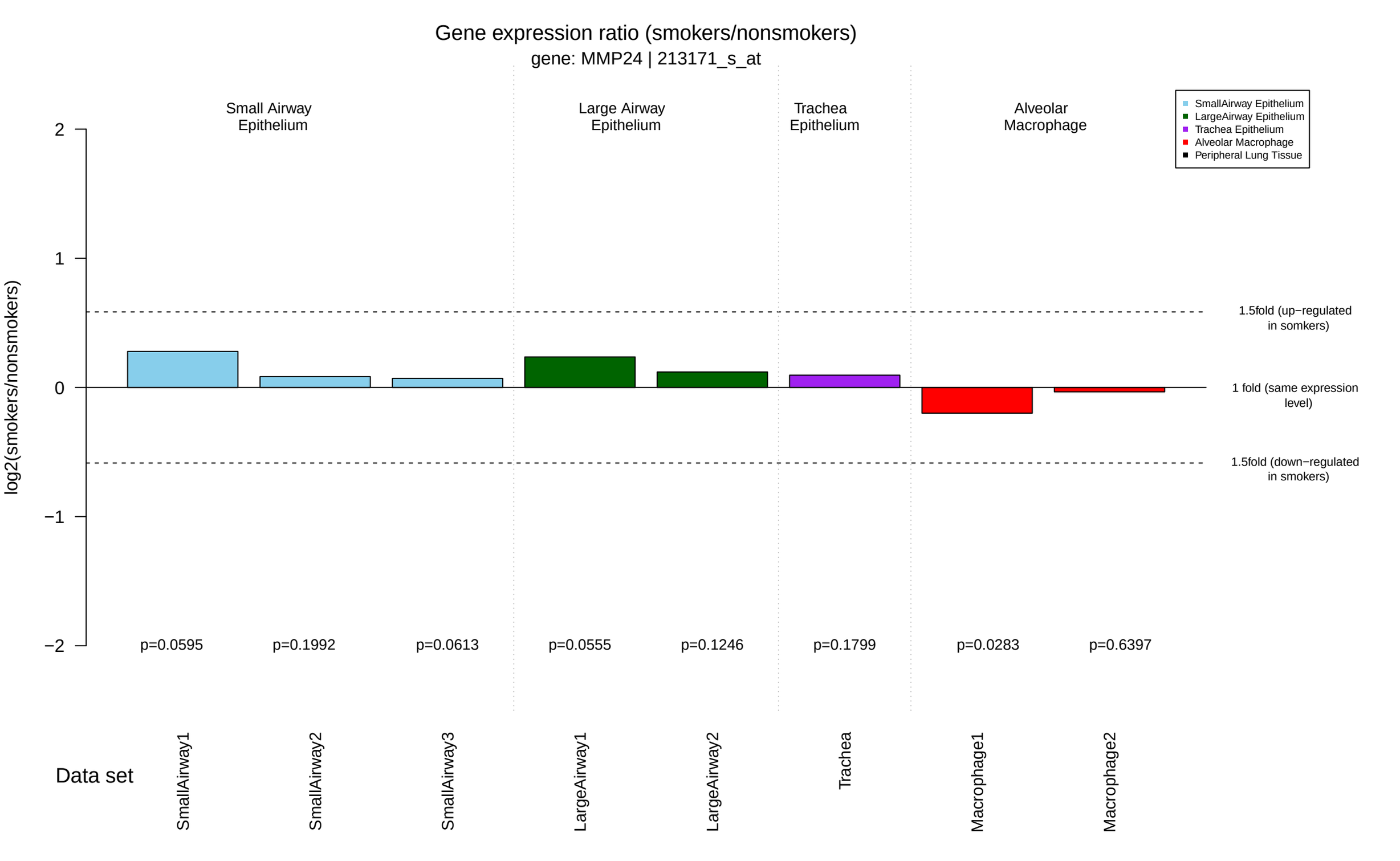
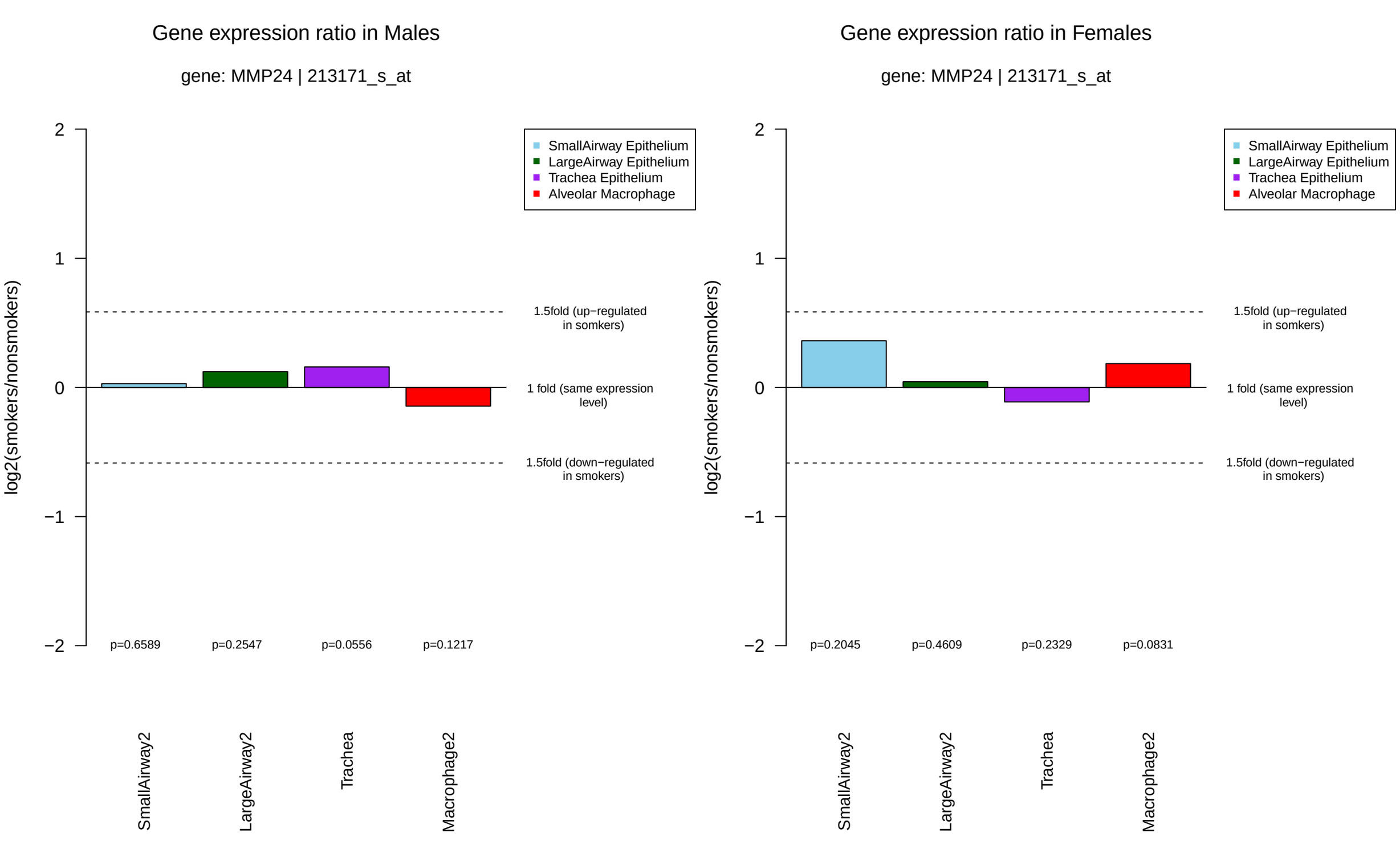
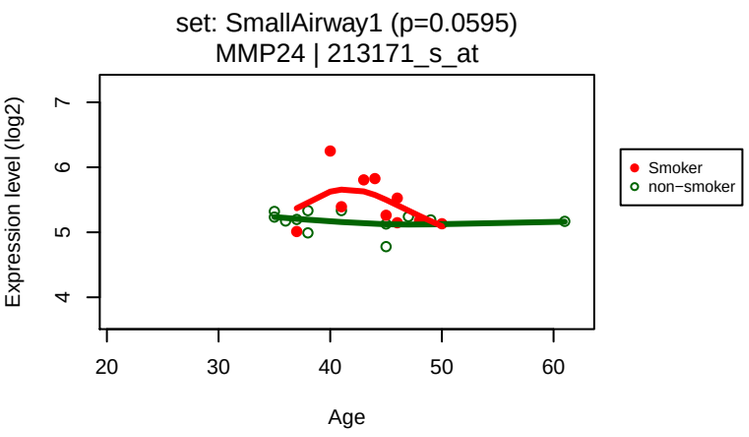
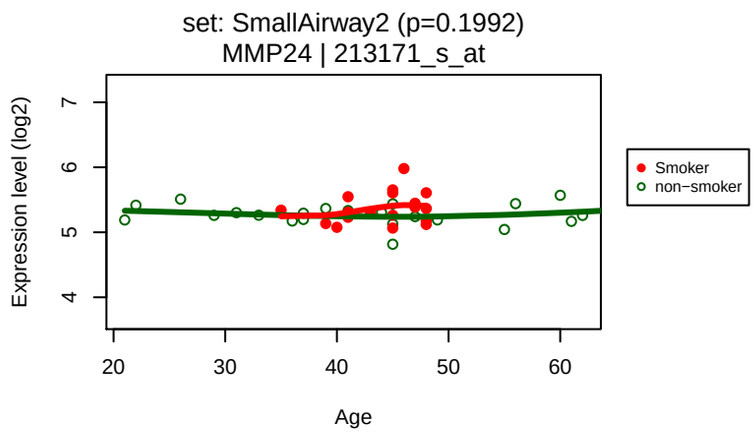
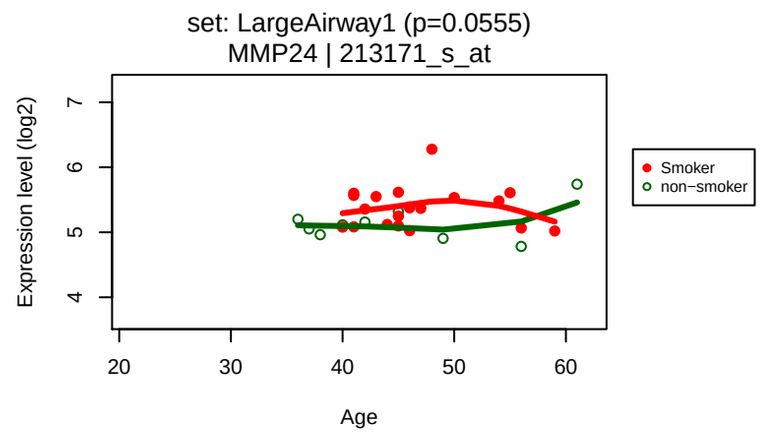

Gene expression of smokers and nonsmokers
gene: MMP24 | 213171_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0595
p=0.1992
p=0.0613
p=0.0555
p=0.1246
p=0.1799
p=0.0283
p=0.6397
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

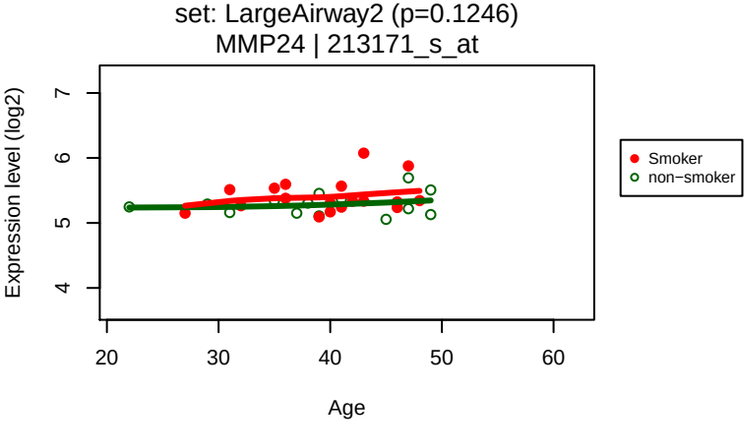

Gene expression ratio (smokers/nonsmokers)
gene: MMP24 | 213171_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0595
p=0.1992
p=0.0613
p=0.0555
p=0.1246
p=0.1799
p=0.0283
p=0.6397
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

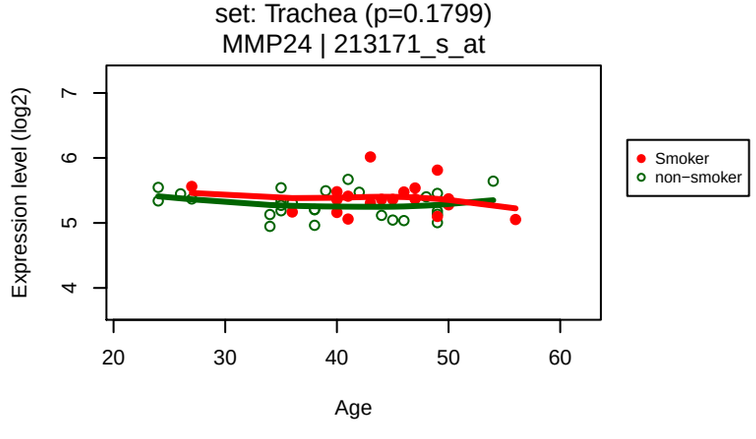

Gene expression ratio in Males
gene: MMP24 | 213171_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6589
p=0.2547
p=0.0556
p=0.1217
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP24 | 213171_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2045
p=0.4609
p=0.2329
p=0.0831
SmallAirway2
LargeAirway2
Trachea
Macrophage2

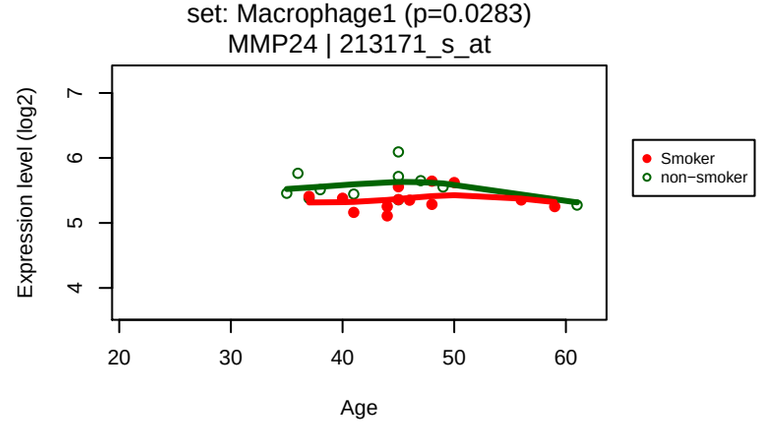

set: SmallAirway1 (p=0.0595)
MMP24 | 213171_s_at
Expression level (log2)
Age
Smoker
non−smoker

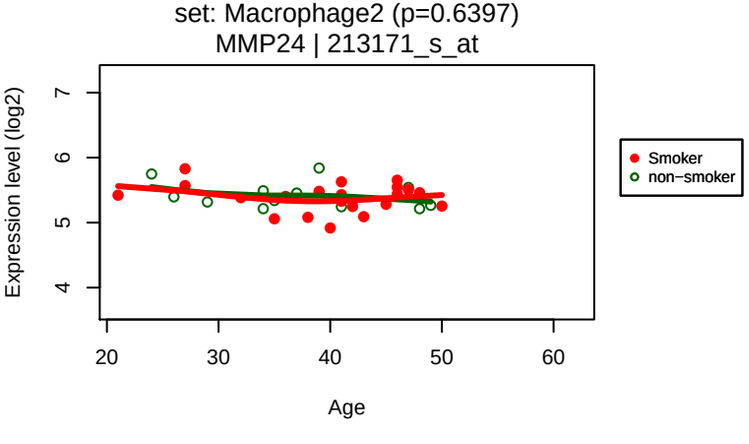

set: SmallAirway2 (p=0.1992)
MMP24 | 213171_s_at
Expression level (log2)
Age
Smoker
non−smoker

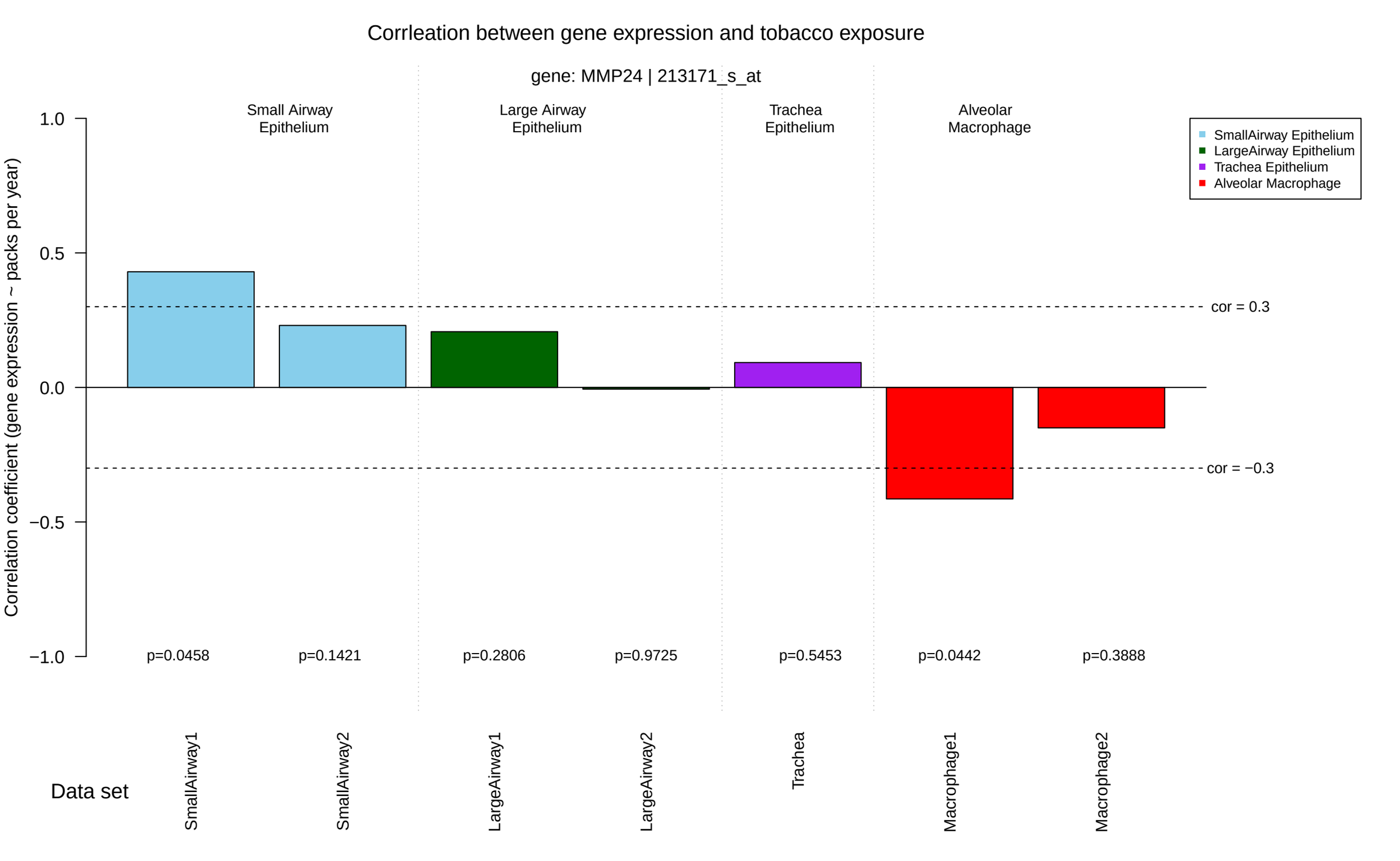

set: LargeAirway1 (p=0.0555)
MMP24 | 213171_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1246)
MMP24 | 213171_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1799)
MMP24 | 213171_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0283)
MMP24 | 213171_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6397)
MMP24 | 213171_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP24 | 213171_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0458
p=0.1421
p=0.2806
p=0.9725
p=0.5453
p=0.0442
p=0.3888
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2